ModelIt!

Teacher's Guide

How Does 5G Actually Work?

Modeling Electromagnetic Wave Behavior in Wireless Communication Networks

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-PS4-2, HS-PS4-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS4-2, HS-PS4-3

Evaluate questions about the advantages of using digital transmission and storage of information. Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model.

Three-Dimensional Learning

Science Practice	**Evaluating Claims and Evidence** — Students evaluate claims about 5G performance by building computational models that reveal the physics-based trade-offs between frequency, penetration, and coverage.
Disciplinary Core Idea	**PS4.A Wave Properties / PS4.C Information Technologies and Instrumentation** — Electromagnetic waves transport energy and information; wave frequency determines both data capacity and interaction with matter, creating fundamental engineering trade-offs.
Crosscutting Concept	**Structure and Function** — Students analyze how the physical structure of electromagnetic waves (frequency, wavelength) determines their functional properties (data capacity, penetration, absorption) and constrains network design.

CAST Testing Alignment

MULTIPLE CHOICE:

A 5G tower using 28 GHz millimeter wave covers approximately 200 meters, while a 4G tower at 700 MHz covers 5 kilometers. Based on your model, how many 5G small cells would be needed to replace one 4G tower's coverage area?

CONSTRUCTED RESPONSE:

Using your model, explain why 5G promises much faster speeds than 4G but requires dramatically more cell towers. Reference the relationship between Signal Frequency, Wave Penetration, Data Transfer Rate, and Tower Density in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every time you stream a video, send a text, or scroll social media, you're using electromagnetic waves traveling at the speed of light between your device and a cell tower. The evolution from 4G to 5G represents a fundamental shift in which electromagnetic frequencies we use — and understanding why higher frequencies mean faster speeds but shorter range reveals the core physics of wireless communication.

The Electromagnetic Spectrum and Data

All wireless communication uses electromagnetic waves — the same fundamental phenomenon as visible light, radio waves, and X-rays, differing only in frequency. Data is encoded by modulating the wave's amplitude, frequency, or phase. The amount of data a wave can carry is directly proportional to its bandwidth (the range of frequencies used). Higher carrier frequencies allow wider bandwidths. A 28 GHz millimeter wave channel can be 400 MHz wide (carrying 10+ Gbps), while a 700 MHz 4G channel is typically 20 MHz wide (carrying 100 Mbps). This is why 5G is faster — it uses higher frequencies with wider channels.

The Penetration Problem

Electromagnetic waves interact with matter in frequency-dependent ways. Low-frequency waves (700 MHz, wavelength ~43 cm) can diffract around obstacles and penetrate building materials with moderate loss. High-frequency millimeter waves (28 GHz, wavelength ~1 cm) are absorbed and reflected by most solid materials. A 28 GHz signal loses approximately 40 dB passing through a concrete wall — that's 99.99% energy loss. Even glass windows, tree foliage, and rain attenuate millimeter waves significantly. This is a consequence of the wave's wavelength being similar in size to common obstacles, maximizing interaction.

Small Cells and Densification

Because millimeter wave signals cover only 200-300 meters (versus 2-5 km for 4G), 5G requires network densification — deploying many small cell antennas on light poles, building facades, and utility poles rather than relying on a few large towers. A single city block might need 4-6 small cells for millimeter wave coverage. This creates challenges: more antennas mean more interference between cells, more power consumption, more installation costs, and more community resistance to antenna proliferation.

The Layered Solution

Real 5G networks don't use one frequency — they layer three bands. Low-band (600-900 MHz) provides wide coverage similar to 4G with modest speed improvements. Mid-band (2.5-6 GHz) offers a balance of coverage and speed for urban areas — this is where most users experience 5G. Millimeter wave (24-100 GHz) provides extreme speed in dense locations like stadiums, airports, and downtown areas. Your phone automatically switches between bands based on availability. This layered approach is an engineering compromise that maximizes the strengths of each frequency range.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the 5G network system: Signal Frequency, Wave Penetration, Data Transfer Rate, Tower Density, Interference Level, Energy Absorption, and Network Capacity — with Signal Frequency and Tower Density as external engineering choices.

E **Establish Relationships**

Students determine that increasing Signal Frequency dramatically increases Data Transfer Rate but decreases Wave Penetration and increases Energy Absorption. This forces higher Tower Density, which increases Interference Level. Net Network Capacity depends on balancing all these trade-offs.

V **Visualize & Model**

Students build a computational model showing how frequency selection cascades through all network parameters, revealing the fundamental physics trade-off in wireless communication.

E **Evaluate Outcomes**

Students run 4G standard, 5G millimeter wave, and 5G mid-band scenarios to compare data speed, coverage area, and infrastructure requirements for each frequency strategy.

R **Revise & Extend**

Students add Beamforming Precision, Weather Impact, or Latency to explore how advanced antenna technology and environmental conditions affect network performance.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"5G promises to download a movie in three seconds. But walk behind a wall and the signal vanishes. Why? The answer is pure physics."*

Watch For

Demo: If possible, compare signal strength between 4G and 5G in the classroom using a signal strength app. If not, show a signal propagation comparison video.

Slide 2: Learning Objectives

3 min

Say: *"Today you're learning the physics behind every text, every stream, every download — and why engineers had to make painful trade-offs to give you faster internet."*

Watch For

Have students read objectives. Pre-teach 'millimeter wave' and 'signal attenuation.' Quick-write: Why do you think 5G is faster than 4G?

Slide 3: Big Question

3 min

Say: *"You'd think faster internet is just about better technology. But it's actually about physics — and physics doesn't negotiate."*

Watch For

Students guess: What happens when you double the frequency of a wave? What happens to its ability to go through walls? Collect hypotheses.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the electromagnetic physics behind wireless networks — and discover why engineers have to compromise between speed and coverage."*

Watch For

Preview each LEVER step. Emphasize that this is about the physics of electromagnetic waves interacting with matter.

Slide 5: Activity 1: Components

8 min

Say: *"Two things the engineers choose: what frequency to broadcast on and how many towers to build. Physics determines the rest."*

Watch For

Guide component sorting. Discuss why Signal Frequency and Tower Density are external (engineering choices) while Wave Penetration and Energy Absorption are governed by physics.

Slide 6: Activity 2: Connections

8 min

Say: *"When you crank up the frequency for more speed, what chain reaction ripples through every other component of the network?"*

Watch For

Help students trace the trade-off cascade: higher frequency means more data but less penetration, requiring more towers, creating more interference. Find the balance points.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare 4G, 5G mid-band, and 5G millimeter wave. Same city, same users, completely different physics."*

Watch For

Students predict outcomes for each frequency band. Run simulations. Compare tower counts, coverage gaps, and data speeds side by side.

Slide 8: Discoveries

5 min

Say: *"Now you know why there's no perfect wireless solution — only trade-offs. The real genius of 5G is using ALL the frequencies and switching between them."*

Watch For

Lead discussion on the layered network approach. Connect model findings to students' real experience with cell signal variability.

Slide 9: STEM Challenge

12 min

Say: *"Your school just got funding for a 5G network. Design it. Where do the antennas go? What frequencies do you use? And what happens during the school assembly?"*

Watch For

Groups design campus networks using model data. Must specify antenna locations, frequency bands, and capacity for high-density events. Present as engineering proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Signal Frequency, Tower Density
- **INTERNAL:** Wave Penetration, Data Transfer Rate, Interference Level, Energy Absorption, Network Capacity

Relationships

- **Signal Frequency to Data Transfer Rate:** POSITIVE (+) — Higher Signal Frequency allows wider bandwidth channels, which can carry proportionally more data per second. Doubling the frequency roughly doubles the available bandwidth and data capacity.
- **Signal Frequency to Wave Penetration:** NEGATIVE (-) — Higher Signal Frequency produces shorter wavelengths that interact more strongly with building materials, vegetation, and atmospheric particles. Millimeter waves at 28 GHz lose 99.99% of energy through a concrete wall, while 700 MHz signals pass with moderate attenuation.
- **Tower Density to Interference Level:** POSITIVE (+) — Higher Tower Density means more overlapping signals in the same geographic area. Without sophisticated interference management (beamforming, MIMO), signals from adjacent cells degrade each other, reducing effective Data Transfer Rate and Network Capacity.

Key Discoveries

- There is a fundamental physics trade-off in wireless communication: higher frequencies carry more data but penetrate materials less effectively and are absorbed more by the atmosphere
- 5G millimeter wave delivers extraordinary data rates (10+ Gbps) but requires a massive increase in Tower Density because each cell covers only 200-300 meters versus 2-5 kilometers for 4G
- Interference Level increases with Tower Density, requiring sophisticated beamforming and MIMO technology to direct signals precisely and manage overlapping coverage
- The 'best' 5G solution is actually a layered network using multiple frequency bands — low-band for wide coverage, mid-band for urban areas, and millimeter wave for dense hotspots

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"5G is dangerous to human health"	5G uses non-ionizing electromagnetic radiation — the same fundamental type as FM radio, Wi-Fi, and visible light. Non-ionizing radiation does not have enough energy per photon to break chemical bonds or damage DNA. The frequencies used by 5G (up to 39 GHz) are far below the ionizing threshold (approximately 3,000,000 GHz for UV light). Decades of research have found no confirmed health effects from non-ionizing radiation at the power levels used by cell networks.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"5G is just faster 4G"	5G is not simply 4G with higher speeds — it represents a fundamental architectural change in wireless networks. 5G introduces network slicing (dedicating virtual network segments to different applications), massive MIMO (hundreds of antennas per tower), edge computing (processing data near the user rather than in distant data centers), and ultra-low latency (1ms vs 30-50ms). These capabilities enable entirely new applications like autonomous vehicles, remote surgery, and industrial automation that were impossible on 4G.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Higher frequency always means better performance"	The model demonstrates that higher frequency increases data rate but simultaneously decreases coverage area, increases infrastructure costs, and introduces vulnerability to weather and obstacles. There is no universally 'better' frequency — each range offers different trade-offs. The optimal frequency depends on the specific use case: low-band for rural coverage, mid-band for suburban areas, and millimeter wave for dense urban hotspots.